BFW STEAM HEATER VER 1

Operator Training Simulator · E-101

Chulalongkorn University · Department of Chemical Engineering

DOCUMENT REGISTER AND ENGINEERING QUERY LOG

DOCUMENT NO. : BFWH-V1-PM-DOC-001

DISCIPLINE : Project (document control)

PROJECT : BFW Steam Heater Ver 1 · E-101 Operator Training Simulator

STATUS : Issued for teaching (IFT)

Register of the 14 documents of the BFW Steam Heater Ver 1 simulator known at this issue (8 from the Engineering (EPC) tabs), the 37 engineering queries (32 open) and the closing document of each of the 28 HAZOP recommendations. The Project & deliverables tab of the simulator holds the live version.

Rev	Date	Description	Originator	Checker	Approver
0	25-Oct-26	Issued for teaching	A. Soottitantawat	—	—

Numbers are taken from the BFW Steam Heater Ver 1 simulator model; values marked assumed / TBC / HOLD are not design data.

CONTENTS

1 How the project is organised 2

2 Document register 3

3 Deliverables by discipline 4

4 Engineering query log 5

5 HAZOP recommendations: owner and closing document 8

6 Register checks at this issue 10

Notes 10

1 How the project is organised

Engineering is split into disciplines; each writes its own documents from the documents of the discipline before it. Process fixes what the unit must do; Mechanical, Instrumentation and Electrical turn that into equipment that can be bought; Safety reviews the design and sends recommendations back; Commissioning turns the documents into tests and a start-up. Every document has a number, a revision and a status in this register. When two documents disagree or a value is missing, an engineering query is raised instead of a guess.

Numbering: BFWH-V1-<discipline>-<type>-nnn. Discipline codes: PE Process, ME Mechanical, PI Piping, IN Instrumentation, EE Electrical, SA Safety, CM Commissioning, PM Project, TR Training. Legacy numbers P&ID-001 and BFW-PFD-100 are kept until their next revision (EQ-04).

Type	What it is for	Written by	Used by
PFD	Shows the process, the main equipment and the heat and material balance at design	Process	All disciplines; the basis of every other document
PID	Shows every line, valve, instrument and safety device: the master drawing of the unit	Process, with Instrumentation and Piping	Construction, commissioning, operations, HAZOP
PE-HMB	Heat and material balance: flows, temperatures, pressures of each stream	Process	Mechanical, Instrumentation (ranges), Electrical (pump duty)
PE-RLB	Relief load basis: every overpressure case and the relief device size	Process (relief)	Mechanical (nozzles), Piping (discharge), Commissioning (PSV test)
PE	Process data: what the unit must do and the conditions each item sees	Process	Mechanical, Instrumentation, Electrical
PI-LAY	Plot plan, elevations and piping isometrics: where each item stands and how each line is routed and built	Piping (layout)	Civil (foundations), construction, 3-D model, operations (access)
ME	Mechanical design and data sheets: what to buy and how it is built and tested	Mechanical	Procurement, vendors, inspection
IN-CE	Cause and effect: which trip cause gives which action	Instrumentation	SIS programmer, commissioning (proof tests), operations
IN-SRS	Safety requirements of each SIF: set point, response time, SIL, proof test	Instrumentation / Process safety	SIS design, proof tests, operations
IN	Instrumentation: instrument index, data sheets, control and alarm philosophy	Instrumentation	Procurement, DCS/SIS configuration, loop checks
SA-HZP	HAZOP report: deviations, safeguards, risk ranking and recommendations	Process safety (HAZOP team)	Every discipline closes its recommendations
SA	Safety studies and their actions	Process safety	All disciplines
EE-LL	Electrical load list: kW and kVA per consumer, maximum demand	Electrical	Transformer / MCC / UPS sizing, procurement
EE	Electrical design: supply, motors, cables, area classification	Electrical	Procurement, construction
CM-PLN	Plan from purchase to handover: packages, tests, checklists, first start-up	Commissioning	Procurement, construction, commissioning, operations
PM-DOC	Document register and engineering query log: what exists, at which revision, and what is open	Project (document control)	Everyone on the project
TR	Training material for operators and instructors	Training	Instructors, trainees

2 Document register

Training material for instructors is registered in the instructor edition of the simulator page only and is not listed here.

Document no.	Title	Discipline	Rev	Pages	Tab	File	Written by	Used by	Note
BFWH-V1-PM-DOC-001	Document register and engineering query log	Project	0	10	Project & deliverables	BFWH-V1-PM-DOC-001.pdf	Project (document control)	Everyone on the project	
BFW-PFD-100	Process flow diagram with stream table	Process	0	1	PFD	bfw-pfd-100.pdf	Process	All disciplines; the basis of every other document	Legacy number; A4 sheet.
BFWH-V1-PE-HMB-001	Heat and material balance and line sizing	Process	0	8	Process engineering	BFWH-V1-PE-HMB-001.pdf	Process	Mechanical, Instrumentation (ranges), Electrical (pump duty)	
BFWH-V1-PE-RLB-001	Relief load basis and relief device sizing	Process	0	8	Relief & vent	BFWH-V1-PE-RLB-001.pdf	Process (relief)	Mechanical (nozzles), Piping (discharge), Commissioning (PSV test)	
BFWH-V1-PID-001	Piping and instrumentation diagram, E-101 BFW steam heater	Process	D	1	P&ID	BFWH-V1-PID-001.pdf	Process, with Instrumentation and Piping	Construction, commissioning, operations, HAZOP	A3. Line numbers, valves, instruments, notes; sizes TBC.
P&ID-001	P&ID, operating screen drawing	Process	C	—	P&ID	no PDF	Process, with Instrumentation and Piping	Construction, commissioning, operations, HAZOP	Legacy number; the drawing behind the DCS / P&ID screens. Superseded for engineering by BFWH-V1-PID-001 rev D.
BFWH-V1-ME-EQL-001	Equipment list and datasheets	Mechanical	0	8	Mechanical	BFWH-V1-ME-EQL-001.pdf	Mechanical	Procurement, vendors, inspection	
BFWH-V1-PI-LAY-001	Plot plan, elevations and piping isometrics	Piping	0	15	Layout & isometrics	BFWH-V1-PI-LAY-001.pdf	Piping (layout)	Civil (foundations), construction, 3-D model, operations (access)	
BFWH-V1-IN-CE-001	Cause and effect diagram, SIS I-100	Instrumentation	0	2	C&E matrix	BFWH-V1-IN-CE-001.pdf	Instrumentation	SIS programmer, commissioning (proof tests), operations	A3.
BFWH-V1-IN-INL-001	Instrument list and I/O schedule	Instrumentation	0	7	Instrumentation & control	BFWH-V1-IN-INL-001.pdf	Instrumentation	Procurement, DCS/SIS configuration, loop checks	
BFWH-V1-IN-SRS-001	Safety requirements specification of the SIFs	Instrumentation	R0	—	SRS of SIF	no PDF	Instrumentation / Process safety	SIS design, proof tests, operations	Issued on the SRS tab (CSV), no PDF.
BFWH-V1-EE-LL-001	Electrical load list	Electrical	0	5	Electrical	BFWH-V1-EE-LL-001.pdf	Electrical	Transformer / MCC / UPS sizing, procurement	
BFWH-V1-SA-HZP-001	HAZOP study report	Safety	0	55	HAZOP	BFWH-V1-SA-HZP-001.pdf	Process safety (HAZOP team)	Every discipline closes its recommendations	Nodes N1–N6, recommendations R01–R28.
BFWH-V1-CM-PLN-001	Procurement, completion and commissioning plan	Commissioning	0	8	Procurement to commissioning	BFWH-V1-CM-PLN-001.pdf	Commissioning	Procurement, construction, commissioning, operations	

3 Deliverables by discipline

Discipline	Documents	Numbers
Project	1	BFWH-V1-PM-DOC-001
Process	5	BFW-PFD-100, BFWH-V1-PE-HMB-001, BFWH-V1-PE-RLB-001, BFWH-V1-PID-001, P&ID-001
Mechanical	1	BFWH-V1-ME-EQL-001
Piping	1	BFWH-V1-PI-LAY-001
Instrumentation	3	BFWH-V1-IN-CE-001, BFWH-V1-IN-INL-001, BFWH-V1-IN-SRS-001
Electrical	1	BFWH-V1-EE-LL-001
Safety	1	BFWH-V1-SA-HZP-001
Commissioning	1	BFWH-V1-CM-PLN-001

4 Engineering query log

No.	Subject	Query	Raised by	Refers to	Tab	Status
EQ-01	One shell design pressure for E-101	The Design tab uses a shell design pressure of 350 kPa g (= PSV-100 set) for the relief sizing; the LOPA and Safety layers tabs and the LOPA deck use 300 kPa g. Every document that quotes it must carry one value: the relief basis, the hydrotest pressure (455 vs 390 kPa g), the SRS hazard of PSHH-100 and the mechanical data sheet. Linked: EQ-30 (relief), EQ-60 (hydrotest).	Project	BFWH-V1-SA-HZP-001 (R08)	Project & deliverables	Open
EQ-02	PSV-100 set pressure TBC	BFWH-V1-PID-001 rev D shows PSV-100 set TBC. The set fixes the bench test (EQ-61), the relief sizing and the shell nameplate. Close together with EQ-01; linked EQ-33.	Project	BFWH-V1-PID-001	Project & deliverables	Open
EQ-03	FV-101B fail position TBC	The fail position of the minimum-flow recycle valve FV-101B is not shown on the P&ID and is TBC (HAZOP R22). Fail open protects the pump on air failure but takes BFW from E-101; fail closed keeps the flow to E-101 but can dead-head the pump if FV-101A also closes. Decide, show it on the P&ID and in the C&E, and test it at the stroke test.	Project	BFWH-V1-IN-CE-001	Project & deliverables	Open
EQ-04	Legacy numbers P&ID-001 rev C and BFW-PFD-100	Two documents do not follow the BFWH-V1---nnn pattern. Answer: BFWH-V1-PID-001 rev D supersedes P&ID-001 rev C for engineering; rev C stays as the operating-screen drawing. BFW-PFD-100 keeps its number until its next revision, when it is reissued as BFWH-V1-PE-PFD-001.	Project	BFW-PFD-100	Project & deliverables	Closed
EQ-10	1"-VT-101 vent line velocity	At the continuous-vent rate of 1 % of the design steam (99 kg/h, Design tab) the 1" vent line runs at 43.7 m/s, above the typical 40 m/s; with V-104 wide open for purging the line is choked. Confirm the vent-rate basis (rev B RO-101 orifice or rev C AAV-101) and consider 1½"-VT-101 (18.5 m/s).	Process	BFWH-V1-PE-HMB-001 / BFWH-V1-PID-001 rev D	Process engineering	Open
EQ-11	4"-SC-102 two-phase velocity after the trap	Condensate flashes 2.24 % at design and 3.66 % at maximum load (shell 158.5 kPa g) when it drops to the 30 kPa g return. Homogeneous velocity in 4"-SC-102 is 10.4 m/s at design but 20.5 m/s at maximum load, above the typical 15 m/s. Confirm 6"-SC-102 (9.1 m/s at maximum load) or accept the higher velocity with an erosion and hammer review; confirm the return-header back-pressure.	Process	BFWH-V1-PE-HMB-001 / BFWH-V1-PID-001 rev D	Process engineering	Open
EQ-12	PDI-109 high alarm below the rated-case tube ΔP	With clean tubes the rated (maximum-load) case gives a tube ΔP of 88.2 kPa, above the PDI-109 H alarm of 80 kPa, so the alarm would stand whenever the unit runs at its rated load. Propose a flow-compensated alarm on $\Delta P/F^2$ or a set point above the rated value; Instrumentation to confirm.	Process	BFWH-V1-PE-HMB-001 / alarm table	Process engineering	Open
EQ-13	TK-101 elevation and minimum level for NPSH	NPSH available (13.0 m) uses the model's suction basis K.Ptk = 30 kPa g as 3.06 m of minimum liquid level above the pump centreline, with an assumed suction line of 25 m equivalent length. Confirm the TK-101 elevation, minimum operating level and suction layout; NPSHr is HOLD for the pump vendor.	Process	BFWH-V1-PE-HMB-001 / BFWH-V1-ME-EQL-001	Process engineering	Open
EQ-20	E-101 shell design pressure 350 vs 300 kPa g	The Design tab sets the shell design pressure at 350 kPa g (= PSV-100 set, assumed) and this equipment list uses it; the LOPA tab (bow-tie BT1) and the Safety layers tab use a shell rated 300 kPa g (assumed). Reconcile to one value. At 300 kPa g the shell design equals the assumed header maximum, leaving no margin for the FV-104-fails-open case at the header maximum (model 267 kPa g) plus PSV accumulation.	Mechanical	BFWH-V1-ME-EQL-001 / LOPA tab / Design tab	Mechanical	Open
EQ-21	Steam trap ST-101 capacity	The datasheet requires 24 838 kg/h at 75 kPa (SF 2.5 on the design condensate, start-up with a cold shell). The trap in the simulator model passes 14 840 kg/h at 75 kPa (1.49 × design). Confirm the trap selection with the vendor, or accept a lower safety factor with the slow warm-up in the start-up procedure; the model then represents a smaller trap than the datasheet.	Mechanical	BFWH-V1-ME-EQL-001 / Design tab	Mechanical	Open
EQ-22	Fixed-tubesheet differential expansion	Shell at 121.3 °C and tube metal at about 92 °C give 3.3 mm differential expansion over 9.39 m at design and 11.6 mm at start-up (tubes at 18 °C, shell hot). A fixed-tubesheet BEM needs a shell expansion joint or a tubesheet design (UHX) that takes the load; alternatives are a floating head or U-tubes (two passes, changes the thermal design). Expansion joint and tubesheet HOLD.	Mechanical	BFWH-V1-ME-EQL-001	Mechanical	Open
EQ-23	E-101 shell full-vacuum rating	After a trip the shell condenses toward the saturation pressure of the BFW (near full vacuum). An indicative long-cylinder check gives 274 kPa allowable for an 8 mm shell, above 101.3 kPa, but the UG-28 calculation and the FV stamp are not done. Confirm the FV rating so protection does not rely on VB-101 alone (HAZOP R14).	Mechanical	BFWH-V1-ME-EQL-001 / HAZOP R14	Mechanical	Open

No.	Subject	Query	Raised by	Refers to	Tab	Status
EQ-24	P-101 motor rating basis differs between Mechanical and Electrical	This equipment list sizes the P-101A/B motor on 1.15 × the absorbed power at the rated point (26.6 kW at 123.6 m³/h) = at least 30.6 kW. The electrical load list (BFWH-V1-EE-LL-001) sizes it on 1.15 × the peak of the absorbed-power curve (27.89 kW near 105 m³/h; maximum-load case 125.6 m³/h) = 32.1 kW and selects a 37 kW motor. Agree one sizing basis (rated point or end of curve / maximum flow, and the margin) and update both documents; the pump datasheet should state the motor rating.	Mechanical	BFWH-V1-ME-EQL-001 / BFWH-V1-EE-LL-001	Mechanical	Open
EQ-30	E-101 shell design pressure	The Design tab sizes PSV-100 with a shell design pressure of 350 kPa g; the LOPA and Safety layers tabs (and the LOPA deck) use 300 kPa g. Confirm one shell design pressure (= PSV-100 set) and align all documents. At 300, the FV-104-open case at the 300 kPa g header maximum (model 267 kPa g) still stays below set, but the margin to PSHH-100 (250) shrinks.	Process (relief)	BFWH-V1-PE-RLB-001 / LOPA tab	Relief & vent	Open
EQ-31	Steam header maximum pressure	The relief basis rests on a header maximum of 300 kPa g (assumed, top of the simulator fault range). With a let-down failure toward 400 kPa g the model shell reaches 367 kPa g and crosses 350 at 8 s; FV-104 wide open would then need 10 552 kg/h of relief (orifice Q). Confirm the header maximum and its protection (pressure control or relief on the header, HAZOP R08/R09) or size PSV-100 for this case.	Process (relief)	BFWH-V1-PE-RLB-001 S2c	Relief & vent	Open
EQ-32	Tube-rupture relief device	Tube rupture governs (22 452 kg/h BFW, two-thirds rule 733.7 > 350 kPa g). PSV-100 sized for steam (orifice G) passes only about 21 050 kg/h of water and is too slow for the pressure wave of a broken tube. Rupture disc RD-100 recommended (HAZOP R11): confirm by transient analysis, set burst pressure and discharge route. HOLD.	Process (relief)	BFWH-V1-PE-RLB-001 / BFWH-V1-PID-001 note 9	Relief & vent	Open
EQ-33	PSV-100 set pressure	BFWH-V1-PID-001 rev D shows PSV-100 'SET TBC'. This document uses 350 kPa g (= shell design, Design tab). Confirm the set pressure from the E-101 datasheet and update the P&ID and the C&E.	Process (relief)	BFWH-V1-PID-001 rev D	Relief & vent	Open
EQ-34	VB-101 size and shell vacuum rating	Replacing the full condensing rate after XV-100 closes (2.12 kg/s at 0 kPa g) needs a breaker bore of about 194 mm at 20 kPa vacuum. Confirm the shell vacuum rating (-30 kPa g assumed on the Safety layers tab) or full-vacuum design, and the VB-101 size and test interval (HAZOP R14).	Process (relief)	BFWH-V1-PE-RLB-001 section 4	Relief & vent	Open
EQ-40	FV-101B fail position	The minimum-flow recycle valve FV-101B has no fail position on BFWH-V1-PID-001 rev D (note 6: TBC) and none in the model. Fail open protects the pump on loss of air but bypasses BFW from E-101 while FV-101A also fails open; confirm the fail position and add it to the C&E (HAZOP R22).	Instrumentation & control	BFWH-V1-IN-INL-001 / P&ID note 6	Instrumentation & control	Open
EQ-41	SIS inputs share BPCS transmitters	FSLL-101 reads FT-101 (the FIC-101 control transmitter) and PSHH-100 reads PT-100 (PI-100). One failed transmitter can cause the upset and blind the trip. Own transmitters FT-101S and PT-100S (tags TBC) wired to the SIS are recommended; HOLD until the SRS / LOPA confirm the SIL credit (P&ID note 7).	Instrumentation & control	BFWH-V1-PID-001 note 7 / SRS tab	Instrumentation & control	Open
EQ-42	FT-101 turndown at the FSLL-101 trip	FT-101 is ranged 0-160 m3/h; the FSLL-101 trip at 30 m3/h is 18.8 % of flow and only 3.5 % of the DP span, below the readable limit of 40 m3/h for an orifice loop (4:1, assumed). Range the own SIS transmitter FT-101S for the trip (e.g. 0-120 m3/h, trip at 25 % flow) or use a meter with a wider turndown.	Instrumentation & control	BFWH-V1-IN-INL-001 section 4	Instrumentation & control	Open
EQ-43	Trip settings near the top of the span	TSHH-108 at 95 C is 92.9 % of the TT-108 span (30-100 C) and PSHH-100 at 250 kPa g is 96.3 % of the PT-100 span (-10 to 260 kPa g); the PSV-100 set (350, assumed) is off scale. Re-range: TT-108 0-150 C, shell pressure -100 to 400 kPa g (compound, also shows vacuum on cooldown).	Instrumentation & control	BFWH-V1-IN-INL-001 section 4	Instrumentation & control	Open
EQ-44	PSHH-100 response time	The SRS tab gives a PSHH-100 response of 5.75 s (3 s delay + transmitter and logic + XV-100 stroke 2 s, assumed) against a process safety time of 5 s. Shorten the trip delay, confirm the XV-100 stroke time on its datasheet, and confirm the transmitter response.	Instrumentation & control	SRS tab / BFWH-V1-IN-SRS-001	Instrumentation & control	Open
EQ-45	XV-100 stroke time and hotwell level	The XV-100 closing time of 2 s is assumed in the model and the SRS; obtain it from the valve datasheet. The E-101 hotwell level is not measured (P&ID note 11): LT-100 with a high alarm below the tube bottom (12 % of shell ID) is recommended (HAZOP R12), HOLD.	Instrumentation & control	BFWH-V1-PID-001 note 11 / HAZOP R12	Instrumentation & control	Open
EQ-50	P-101A/B pump curve and efficiency	The load list uses the model's quadratic pump curve (shut-off 1 030 kPa g) and a constant pump efficiency of 0.70 (assumed). With a constant efficiency the absorbed power peaks at 105 m³/h; on a real curve the efficiency falls past the best point and the power can keep rising to run-out (125.6 m³/h at FV-101A 100 %). Vendor curve with efficiency and NPSHr required to confirm the 37 kW motor.	Electrical	BFWH-V1-EE-LL-001	Electrical	Open

No.	Subject	Query	Raised by	Refers to	Tab	Status
EQ-51	Motor cable route length	Cable sizing uses an 80 m route from the MCC to P-101A/B (assumed). Confirm the MCC location and the cable route on the plot plan; the 25 mm² selection holds up to the length at which the running drop reaches 5 % (see the calculator).	Electrical	BFWH-V1-EE-LL-001	Electrical	Open
EQ-52	Direct-on-line start of a 37 kW motor on the 400 V bus	DOL start at about 6 × In (≈ 400 A, assumed) is taken as acceptable. Confirm the supply transformer rating and short-circuit level allow it without a soft starter (voltage dip at the MCC bus, not only in the cable).	Electrical	BFWH-V1-EE-LL-001	Electrical	Open
EQ-53	Auto-start of P-101B (HAZOP R01)	R01 asks for an auto-start of P-101B on a P-101A trip or low PI-106 pressure. The MCC starter of P-101B needs a remote auto-start input from the DCS and a 'healthy / available' signal back; the starter scheme is HOLD until Instrumentation fixes the logic.	Electrical	BFWH-V1-SA-HZP-001	Electrical	Open
EQ-60	E-101 shell hydrotest pressure	The shell hydrotest is 1.3 × design: 455 kPa g with the Design-tab shell design pressure (350 kPa g) but 390 kPa g with the 300 kPa g basis of the LOPA / Safety layers tab. The test pressure cannot be released to the vendor until the shell design pressure is agreed (see EQ-01).	Commissioning	BFWH-V1-CM-PLN-001	Procurement to commissioning	Open
EQ-61	PSV-100 bench test and RD-100	PSV-100 cannot be bench-tested and sealed while its set point is TBC, and the tube-rupture relief device RD-100 is HOLD. Mechanical completion of the shell-relief system needs both; the relief package PK-05 is on the critical path.	Commissioning	BFWH-V1-PID-001	Procurement to commissioning	Open
EQ-62	XV-100 stroke time	The SRS response-time budget uses an XV-100 stroke of 2 s (assumed, TBC by vendor). Make the 2 s closing time a FAT acceptance criterion of PK-04 and repeat it at the SAT and at each proof test.	Commissioning	BFWH-V1-IN-SRS-001	Procurement to commissioning	Open
EQ-63	Steam-blow acceptance and temporary exhaust	The steam-blow target-plate acceptance and the temporary exhaust route for LS-101 / LS-102 are assumed in this plan. Agree the criteria with the owner and the noise / exclusion zone for the blow before construction ends.	Commissioning	BFWH-V1-CM-PLN-001	Procurement to commissioning	Open
EQ-70	Fall of the 3"-SC-101 condensate line to ST-101	The layout routes 3"-SC-101 straight down from N4 (EL 2.297) to EL 0.500 and then 6.11 m level at EL 0.500 to the trap. The vertical drop gives the gravity head (1.80 m, about 16.6 kPa of condensate), but the level legs have no fall. Typical practice is a fall of at least 1:100 toward the trap (61 mm over 6.11 m) so condensate cannot lie in the line at low load or after a trip. Confirm the fall on the isometric (start the level legs at EL 0.561 at the riser) or accept a level line flooded up to the trap. Resolution: Layout Rev 0b lays 3"-SC-101 with a 1:100 fall (EL 0.565 → 0.535 → 0.500 at ST-101); the page check now passes.	Piping	BFWH-V1-PI-LAY-001 (ISO-3) / BFWH-V1-PID-001	Layout & isometrics	Closed
EQ-71	Headroom over the E-101 platforms	The layout runs pipes over the operating platforms below the 2.1 m clear headroom of typical practice: 14"-LS-102 crosses PLT-1 at EL 3.600 (0.42 m clear above the deck, 1.62 m long), 6"-BW-103 crosses PLT-1 at EL 4.000 (0.92 m clear) and 12"-LS-101 with its FV-104 bypass runs along the PLT-2 walkway at EL 4.600 (1.44 m clear). LS-102 and BW-103 cross PLT-1 between the stair STR-1 and the ladder LAD-1, so a person between them must step over or duck under a pipe to reach either escape. Re-route at the next layout revision: cross the deck at 2.1 m clear or pass under it, and place the steam station beside the walkway instead of over it; then update the 3-D model. Resolution: Layout Rev 0b lowers the steam station to EL 4.0 beside a walkway east of the line (PLT-2 E 22.0–23.6), starts PLT-1 at E 22.0 so 14"-LS-102 drops west of it, and takes 6"-BW-103 straight to the rack (3.8 m clear over PLT-1); the page check finds no pipe below 2.1 m.	Piping	BFWH-V1-PI-LAY-001 / Visual plant 3D tab	Layout & isometrics	Closed
EQ-72	PSV-100 discharge line has no line number	The PSV-100 discharge (drawn 4", EL 3.6 to EL 6.5, open to atmosphere at a safe location) has no line number on BFWH-V1-PID-001 rev D, and neither has the 1" shell drain at V-105. Assign line numbers (for example 4"-VT-102 and 1"-DR-101), size the PSV discharge from the relief load (BFWH-V1-PE-RLB-001, PSV-100 size HOLD), and add a weep hole and the discharge reaction support to the isometric.	Piping	BFWH-V1-PID-001 / BFWH-V1-PE-RLB-001	Layout & isometrics	Open
EQ-73	V-115 (FV-104 bypass) out of reach	Layout Rev 0b puts the FV-104 bypass on the west side of 12"-LS-101 at E 20.8, EL 4.0. V-115 is then 1.2 m in plan from the PLT-2 deck edge (E 22.0) and 4.0 m above grade: it can be operated neither from the walkway (reach 0.75 m, typical practice) nor from grade. V-113 and V-114 on the main line (E 21.45, 0.55 m from the deck edge) are reachable. Bring the bypass within reach: for example run it in the vertical plane above the main line (EL 4.6, 1.6 m above the deck, 0.55 m from the edge), or add a chain wheel (operability of a throttling globe on a chain is poor). The bypass is used to keep steam on E-101 while FV-104 is maintained, so it must be operable. Resolution: Layout Rev 0c moves the FV-104 bypass to the walkway side (E 21.85, V-115 at N 16.6), within reach of PLT-2; the page access check now passes.	Piping	BFWH-V1-PI-LAY-001 (ISO-2) / BFWH-V1-PID-001	Layout & isometrics	Closed

No.	Subject	Query	Raised by	Refers to	Tab	Status
EQ-74	Trap bypass tee off the sloped 3"-SC-101	With the 1:100 fall of Layout Rev 0b, 3"-SC-101 is at EL 0.527 at E 30.0, but the V-103 trap bypass still starts at EL 0.500 there: the upstream tee is 27 mm below the line, so the bypass does not connect in the model and ISO-3 shows it as a loose end. Set the bypass start to the line elevation (EL 0.527) or start the bypass upstream of the first fall; check V-101 (EL 0.520 vs line 0.524) at the same time. Resolution: Layout Rev 0c starts the trap bypass at the SC-101 elevation (EL 0.527; V-101 0.524, Y-101 0.511, V-103 0.514); the connection check now passes and ISO-3 closes.	Piping	BFWH-V1-PI-LAY-001 (ISO-3)	Layout & isometrics	Closed

5 HAZOP recommendations: owner and closing document

Each recommendation of BFWH-V1-SA-HZP-001 is assigned to the discipline in its action party and to the document that will close it; all are open at this revision.

Rec.	Recommendation	Node	Risk	Owner discipline	Closing document	Tab
R01	Auto-start P-101B on a P-101A trip or low discharge pressure (PI-106 transmitter, recommended addition), completing inside the FSLL-101 delay; test the spare start monthly.	N1	12 H	Instrument / Operations	BFWH-V1-IN-INL-001	Instrumentation & control
R02	Add level transmitter LT-101 (recommended addition) on TK-101 with a low-level alarm and a low-low pump trip, and a high-level alarm for overfill.	N1	12 H	Instrument	BFWH-V1-IN-INL-001	Instrumentation & control
R03	Provide a fixed minimum-flow line with restriction orifice from the pump discharge to TK-101, independent of FV-101B, so a blocked outlet downstream of E-101 cannot deadhead the running pump.	N1	6 M	Process	BFWH-V1-PE-HMB-001	Process engineering
R04	Confirm a check valve on each pump discharge and on the BFW line to the boiler house; add them to the inspection plan.	N1/N2	4 L	Mechanical	BFWH-V1-CM-PLN-001	Procurement to commissioning
R05	Fit a thermal relief valve on the E-101 tube side (recommended addition), or lock the procedure so the steam is isolated and the shell cooled before the BFW is blocked in.	N2	8 M	Process	BFWH-V1-PE-RLB-001	Relief & vent
R06	Add a deviation alarm between TT-101 (control) and TIAH-108 (independent) so a transmitter reading low or frozen is seen before the boiler house is affected.	N2	9 M	Instrument	BFWH-V1-IN-INL-001	Instrumentation & control
R07	Agree the interface with the boiler house: shared alarms for FIC-101 LO and TIAH-108 HI, a call-out procedure, and include loss of BFW and hot BFW from E-101 in the boiler-house HAZOP.	N1/N2	12 H	Operations / Process	BFWH-V1-PE-HMB-001	Process engineering
R08	Confirm the E-101 shell design pressure is at least the highest credible header pressure (144 °C saturation at 300 kPa g in the fault library) and that PSV-100 is set at or below it and sized for FV-104 fully open at that header pressure and for the external-fire case.	N3/N4	10 H	Process	BFWH-V1-PE-RLB-001	Relief & vent
R09	Header-pressure response procedure with the upstream flash-steam plant for PI-105 LO/HI; consider a pressure-control station on the header so E-101 never sees more than its design pressure.	N3	9 M	Operations / Process	BFWH-V1-PE-HMB-001	Process engineering
R10	Add FV-104 position feedback ZT-104 (recommended addition) with a deviation alarm between FIC-104 output and valve position (stuck or failed valve).	N3	9 M	Instrument	BFWH-V1-IN-INL-001	Instrumentation & control
R11	Apply the tube-rupture (two-thirds) rule: BFW side up to 1030 kPa g at pump shut-off. If the shell design pressure is below two-thirds of the tube-side design pressure, size PSV-100 for the tube-rupture flow of water and flashing water.	N4	10 H	Process	BFWH-V1-PE-RLB-001	Relief & vent
R12	Add a shell hotwell level transmitter LT-100 (recommended addition) with a high alarm below the tube bottom (12 % of the shell) and a high-high level trip closing XV-100 before steam can meet a flooded shell.	N4/N5	12 H	Instrument / Process	BFWH-V1-IN-INL-001	Instrumentation & control
R13	Put a continuous vent in service (rev B RO-101) or the automatic air vent with vent-line temperature (rev C AAV-101, TI-102, TAL-102): gas blanketing gives no alarm for a long time.	N4	8 M	Process	BFWH-V1-PE-HMB-001	Process engineering
R14	Confirm the shell is rated for full vacuum or that VB-101 is sized for the fastest condensation after a trip; add VB-101 to the preventive-maintenance and test plan.	N4	9 M	Mechanical	BFWH-V1-ME-EQL-001	Mechanical
R15	Trap management: regular trap survey (ultrasonic or temperature), Y-101 cleaning after any maintenance, and confirm ST-101 capacity at low shell differential pressure (start-up and near-vacuum operation).	N5	12 H	Mechanical / Operations	BFWH-V1-ME-EQL-001 + SOP-E101-10	Mechanical

Rec.	Recommendation	Node	Risk	Owner discipline	Closing document	Tab
R16	Replace local gauge PG-110 by a return-header pressure transmitter with a high alarm (recommended addition); route shell drain V-105 to a flash pot or blowdown pit, not to an open sewer.	N5	9 M	Process / Instrument	BFWH-V1-IN-INL-001	Instrumentation & control
R17	Route V-104, PSV-100, V-103 and V-105 discharges to a safe location; fit a silencer on the vent; warning signs and hearing-protection zone around the vent and trap set.	N4/N5	9 M	Mechanical / Safety	BFWH-V1-PE-RLB-001	Relief & vent
R18	Start-up procedure hold points: BFW in AUTO first, vent V-104 open, warm-up with FIC-104 below 10 % (the model gives water hammer at 10 % in a cold shell, none at 5 %) until the shell is warm, vent until dry steam blows, then cascade.	N6	12 H	Operations	BFWH-V1-CM-PLN-001 + SOP-E101-02	Procurement to commissioning
R19	Shutdown procedure: steam off and shell depressured before the BFW is stopped or blocked in; open V-105 only after the shell pressure has fallen; keep V-104 open during lay-up.	N6	8 M	Operations	BFWH-V1-CM-PLN-001 + SOP-E101-03	SOP
R20	Trip-bypass management for FSLL-101, TSHH-108 and PSHH-100: permit, time limit, MOC, bypass status shown on the DCS; proof-test the XV-100 trips at the interval set by LOPA.	N6	8 M	Operations / Instrument	BFWH-V1-IN-SRS-001	Instrumentation & control
R21	Restart after any trip: re-vent the shell (air drawn in through VB-101) before the cascade is put back; consider "V-104 open" as a start permissive.	N6	9 M	Operations / Instrument	BFWH-V1-CM-PLN-001 + SOP-E101-05	SOP
R22	Confirm the fail position of FV-101B (not shown on the P&ID) and that XV-100 closes on loss of air or power; document the fail positions in the cause-and-effect chart.	N6	9 M	Instrument	BFWH-V1-IN-CE-001	Instrumentation & control
R23	LOPA for the High rows that rely on PSHH-100, FSLL-101 and PSV-100 (shell overpressure, tube rupture, loss of BFW, water hammer).	all	10 H	Process safety	BFWH-V1-IN-SRS-001	LOPA
R24	Show a calculated superheat TI-100 − Tsat(PI-100) on the DCS; tell the upstream plant when it exceeds a few degrees.	N3	4 L	Instrument	BFWH-V1-IN-INL-001	Instrumentation & control
R25	Water-treatment control: BFW hardness and oxygen-scavenger sampling programme; closed sampling point and PPE for treatment chemicals.	N1/N2	6 M	Operations	BFWH-V1-PE-HMB-001	Process engineering
R26	Confirm a drip leg and trap on the steam branch upstream of XV-100, and warm the line with its drains open before admitting steam.	N3	8 M	Mechanical / Operations	BFWH-V1-ME-EQL-001	Mechanical
R27	Condensate quality: oil-in-condensate check or monitor on the return to the boiler house; find and stop the oil source in the flash steam.	N4	6 M	Operations	BFWH-V1-PE-HMB-001	Process engineering
R28	Maintenance isolation procedure for E-101: positive isolation (blinds) of steam, BFW and condensate, drain and cool before opening, confined-space permit for shell entry, lift plan for bundle pulls.	N4/N5	9 M	Operations / Maintenance	BFWH-V1-CM-PLN-001 + SOP-E101-11	SOP

6 Register checks at this issue

Check	Result
Document numbers follow BFWH-V1---nnn or the legacy numbers	14 of 14
No duplicate document numbers	none
Every query references a document in the register	37 of 37
Every HAZOP recommendation has an owner document in the register	28 of 28
Shell design pressure: one value in all documents	No — Design tab 350, LOPA / Safety layers 300 kPa g: EQ-01 open
PSV-100 set pressure fixed on BFWH-V1-PID-001	No — SET TBC: EQ-02 open

Notes

- Built by epc/tabs/eproj/doc.py from the docs.json and queries.json of the Engineering tabs present when it ran; the page register is rebuilt every time the simulator opens.
- Page counts of the existing PDFs were read from the issued files; "—" means no PDF or not yet issued.